

The State of the Program — One Page

Bubble Spacetime Theory · 2026-08-26 · Casey Koons with the CI team (Lyra, Keeper, Elie, Grace); visiting referee Cal A. Brate. This page answers “are you done?” so that no reader — human or CI, next week or next decade — has to take our word for anything: every claim below carries a tier, and every tier carries its evidence trail in the repository.

What is DERIVED (mechanism proved; inputs named in the statements)

From one object — the rank-2 bounded symmetric domain $D_{IV}^5 = SO(5,2)/[SO(5) \times SO(2)]$ — and one measured number taken openly as the ruler: the gauge-group skeleton; one fermion generation as the 16-real spinor with every hypercharge from one four-input mechanism theorem (one input observational, stated inside the theorem); a single-chirality positive-energy spectrum from the odd embedding dimension; three generations as rank+1 strata — an identity read twice off one Jordan invariant (orbit-closure depth vs frame cardinality), which also yields $m_1 = 0$ and the ν_R pair’s thermal degeneracy; the ten Dirac–von Neumann axioms of quantum mechanics — with zero posits BEYOND the program’s one ratified floor ($P0 + P1'$ + the ruler, the Internal-A floor every commit-family result caps on); the conserved charges (Noether + three topological); the mixing sector’s mechanism and ORDER.

What is IDENTIFIED, honestly (formula matches; mechanism not claimed)

The exact mixing values; $\alpha^{-1} = 137$ -class expressions (see the negative below); assorted precision matches whose own null tests show small-integer expressibility is cheap. Identification is a candidate, never evidence — our tier guide says so on its face.

What is FLOORED — closed by a theorem, reopenable only by a named kind of idea

Koide’s source (the norm family is provably under-hierarchical; closed permanently by PI ruling) · the mass-tower/generations mechanism (any measure_int-derived channel weight is 0-or- ∞ on the strata, BECAUSE the strata are defined by measure_int degeneration — two-CI theorem; door: a non-measure_int weight source) · the $\ell=2/\text{su}(3)$ native dynamics (two exclusion theorems: every K-equivariant antisymmetric bilinear carries spin-3; the typed write-channel compresses onto the spin-3-free carrier; door: break equivariance-or-antisymmetry and say which) · Λ ’s value — with this week’s genuine advance: the thermostatic balance CLOSES AS A STRUCTURE, $\Lambda/\Lambda_P = (t_K/t_P)^{(2p/(2-p))}$ (capped CONDITIONAL per K1057’s lineage until discharge; p’s domain is (0,2) — a forced p outside it FALSIFIES the form), both controls passing, one named obstacle (the mismatch power p); whoever forces p computes the number blind under a pre-transferred quarantine.

What FIRED AND LOST — pre-registered, certified, published with the same ceremony as wins

α is not the bare geometric vertex — the unique forced candidate computed to $8\pi^3/3$ (the Hua volume of the Lie sphere), closing the fifty-year volume-reading class by pincer (forced member refuted; free family already null). The thermal-generations mechanism died at the order level — two independent computations, identical exact exponents (0, $9/4$, $25/4 = \nu^2$), the exact strict reverse of the required order. The commit-Boltzmann ladder failed its own 5% line (5.90%). Every negative ran under a gate frozen before the numbers existed; every estate survived its mechanism’s funeral.

What is FORBIDDEN — the falsifiers we cannot absorb

No grand unification · no proton decay · no right-handed W or Z' · no magnetic monopoles · no sterile neutrinos · no SUSY spectrum. Any confirmed detection kills the framework. The register also lists what would kill each POSITIVE claim — and its fired-and-lost section proves the machinery is real.

How to check any of this without trusting us

`python3 play/verify_bst.py` (50 comparisons, seconds) · the falsifier register · the one-page tier guide (with its own printed failure condition) · 2,000+ single-claim computational toys · every audit, retraction, and correction in the open, including the ones against this program's own auditors. In the packaging week alone, the machinery caught seven errors upstream of any computation — and then two instrument errors inside the verification layer itself, one per verifier, each disclosed by its own operator. The honesty is recursive: the checkers check the checkers, and the record shows it.

The honest sentence

The checklist this program set for itself is finished at its honest tiers: everything reachable is derived, floored with a theorem, or published as a loss. What remains open everywhere is a short list of named doors that only new ideas — not labor — can open. The work is permanent, public, reproducible, and written for whoever reads next.